

Investigating Interaction Behaviors of Learners in VR Learning Environment

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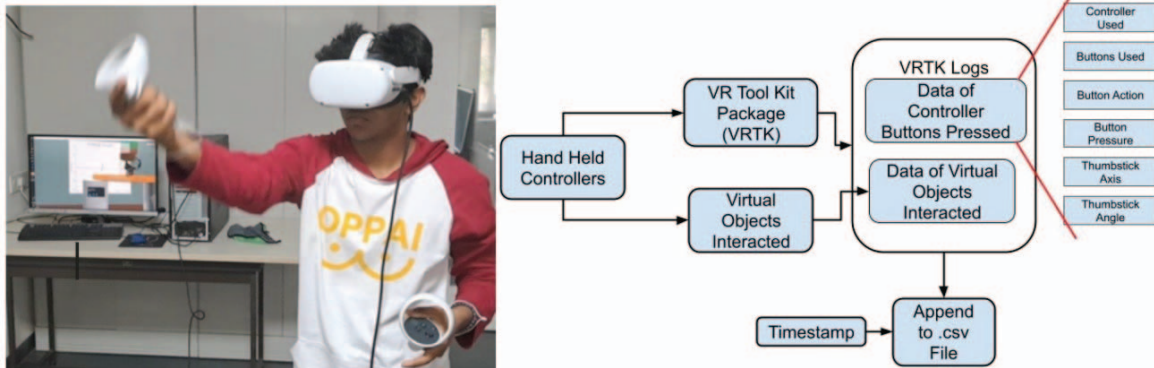
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Figure 1: Learner Interacting in a VRLE (Left) and the Interaction Behavioral Data Collection Mechanism(Right)

ABSTRACT

The education researchers use self-reported questionnaires, physiological sensors, and human observers to study the effects of VR in education. The above-mentioned existing data collection mechanisms do not provide information related to the interaction behavior of the learners in VR Learning Environments (VRLE). As a result, understanding the learning processes from the perspective of learners' behavior in VRLE is still in its infancy. Hence, we developed a mechanism that is able to automatically log all the interaction behavior of the learners happening in VRLE along with timestamps. In this paper, we discuss the study conducted with 14 undergraduate engineering students after deploying the developed real-time Interaction Behavioral Data (IBD) collection mechanism in a room-scale VRLE, MaroonVR. The IBD collected will be analyzed to understand the VR behavior of the learners and the learning processes happening in VRLE. The results of the study could be further utilized to model the learners' behavior in VR and to provide adaptive personalized VR learning content.

Index Terms: Interaction Behavior—Virtual Reality Learning Environment—learner modeling—learning processes

1 INTRODUCTION

Virtual Reality (VR) is a technology that immerses users in the virtual world and makes them interact with virtual objects to give them a more real experience [7]. The distinctive characteristics of VR such as immersion, interaction, and imagination [5] have made the technology be used in various domains such as the Automotive industry, Military, Healthcare, Sports domain, etc including the education domain. There is an exponential increase in the number of studies being conducted to analyze the use of VR in education in the last decade. Most of the existing studies were conducted

by collecting data from pre-tests and post-tests, from self-reported questionnaires, interviews, and surveys [5], and from the devices such as 1) physiological sensors [1], 2) eye trackers [7] 3) body trackers, and 4) orientation of the head-mounted displays (HMD) and handheld controllers (HHC). The data are generally collected to evaluate the user experience, engagement, and usability, and to study the impact of VR by measuring the cognitive, affective, and procedural learning outcomes. In addition, human observers were also employed to collect data related to the affective state [1] and the behavior of the learners. However, the data provided by human observers are biased due to cognitive, social, and communicative causes, and also need to satisfy inter-rater reliability tests in order to become valid. Although the existing studies measured the learning gain and explored learning processes in non-immersive VR systems considering the cognitive and affective factors [2], the knowledge of how learners interact with immersive VRLE and how they learn from it is not explored from the aspect of interaction behavior. This is mainly due to the non-existence of a data collection mechanism that can log the learners' interaction behavior in VRLE. To address this gap, we developed a mechanism that is able to log all the interaction behavior of the learners in VRLE in real time along with the time stamp.

In this paper, we describe the study conducted on 14 participants by deploying the developed data collection mechanism in MaroonVR [3], a VRLE to learn the physics notion of electromagnetic induction. The study result showed that the VR intervention led to an increase in learning gain. The raw data collected in the data logger using the interaction behavioral data (IBD) collection mechanism (refer Figure 1) will be analyzed to extract the temporal and contextual features such as frequency of interactions, duration of interactions, and virtual scenes encountering interactions to assess their impact on the learning outcome.

2 INTERACTION BEHAVIORAL DATA COLLECTION MECHANISM

The IBD collection mechanism is shown in Figure 1. We use the Oculus Quest 2 VR system from Meta in our study. The VR system consists of an HMD to view the VRLE and two HHCs, one for each hand to interact with VRLE. We used the Unity game engine to

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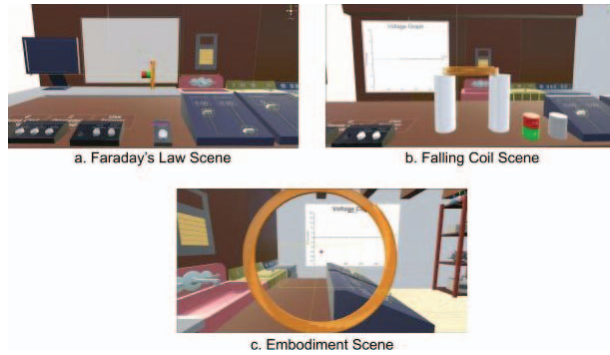


Figure 2: Scenes in VRLE

develop our data collection mechanism [4]. We employed the Virtual Reality Tool Kit (VRTK) package of Unity to collect information related to HHC button actions and wrote a *c#* (say *c sharp*) code in Unity to obtain information related to the virtual objects being interacted with using HHCs. We also wrote another *c#* code to log all the information as a .csv file in a separate folder in the desktop computer to which the HMD is tethered. The *c#* code is written in such a way that, it is able to record time in a separate column when new entries related to interactive actions are appended in the .csv file.

3 VRLE

We adopted MaroonVR [3], an open-sourced VRLE containing conceptual learning content that supports rich interactions similar to procedural learning content and cognitive skill acquisition similar to factual learning content. We adopted two scenes of MaroonVR viz 1) Faraday's law scene and 2) the Falling coil experiment scene. These two scenes are used to learn the notion of electromagnetic induction. Electromagnetic induction is a phenomenon in which when a magnet is moved through the coil, an electromotive force (emf) is induced across the coil. The strength of the induced emf varies as the number of turns in the coil, the area enclosed by the coil, the material (resistance) of the coil, and the magnetic field strength of the magnet differ. Hence, to enhance interactions, we modified the actual MaroonVR by including provisions to change the parameters that affect emf. We also converted the simulated falling coil experiment scene into an interactive scene. Additionally, we included a third scene to the VRLE in which the learners take the perspective of the magnet i.e the learners become the magnet, and based on their movement the induced emf changes. This helps to boost embodiment learning. Figure 2 shows different scenes present in the VRLE.

4 STUDY DESIGN

We deployed the IBD collection mechanism in MaroonVR and conducted a study with 14 engineering undergraduate students. Prior to VR intervention the data related to participants' self-efficacy, self-regulation, and prior knowledge of the phenomena of electromagnetic induction (pre-test) are collected. During VR intervention, the behavioral data are automatically logged using the developed data collection mechanism. After VR intervention the data related to the learning outcome (post-test), learner experience, engagement, enjoyment, and embodiment are collected. But, for this study, the user's self-reported data collected from self-efficacy, self-regulation, experience, engagement, enjoyment, and embodiment questionnaires are not analyzed.

5 RESULTS

The study result showed an increase in normalized learning gain of 0.42 after VR intervention. We also measured the effect size using cohen's *d* as 0.81. As the experiment was conducted with a smaller sample size of 14 participants we performed the Shapiro-Wilk normality test and found that both the pre-test scores and the post-test scores were normally distributed for a significance level of $\alpha = 0.05$ ($w=0.9$, $p=0.112$, and $w=0.908$, $p=0.148$ respectively). Hence, a paired t-test was conducted to test whether the obtained learning outcome was due to VR intervention. From the paired t-test the post-test score was found to be significantly higher than the pre-test score with $t(13)=3.2$, $p=0.04$. The study results are reported in Table 1.

Table 1: Pre-to-post learning gains - all students ($n=14$)

Pre-test score mean (sd)	Post-test score mean (sd)	Normalized Learning Gains	Effect Size Cohen's <i>d</i>	Paired t-test (p-value)
5.86 (1.75)	7.64 (2.06)	0.42 (0.68)	0.81	3.2 (0.04)

6 DISCUSSION AND CONCLUSION

The results of the experiment evidently prove that the VRLE experience with MaroonVR has reported a positive learning outcome. The result of the paired t-test has confirmed that the increase in the learning gain is due to the VR intervention. Also, the integration of the IBD collection mechanism with the VRLE has resulted in the successful logging of learners' interactions. The temporal and contextual features will be extracted from the raw data collected to understand the relationship between the learners' behavior in the VRLE and their performance and hence the learning processes [6]. However, experiments with larger sample sizes need to be conducted to make the result more predictable, generalizable, and transportable. We also propose that the behavioral pattern associated with the performance of the learners can be used to design VR learner models and in turn VR-based Adaptive Tutoring Systems.

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